

28. Orkhan Khalilzade

A model initially estimates that there are 24,000 bacteria on a petri dish. 14 days later, the model estimates that there are 96,000 bacteria on the petri dish. Assuming exponential growth, the formula $B = a(2)^{xd}$ gives the estimated number of bacteria on a petri dish, where a and x are constants and B is the number of bacteria on the petri dish d days after the initial measurement. What is the value of x ?

- A) $\frac{1}{14}$
- B) $\frac{1}{7}$
- C) 7
- D) 14

29. Orkhan Khalilzade

How many liters of a 30% chlorine solution must be added to 12 liters of a 10% chlorine solution to obtain a 15% chlorine solution?

30. Orkhan Khalilzade

In the xy -plane, there are 3 points a , b , and c . Point a has coordinates of $(1, 0)$, point b has coordinates of $(0, 0)$ and point c has coordinates of $(-1, 0)$. Which of the following gives a possible angle measure, in radians of $\angle abc$?

- A) $\frac{456\pi}{6}$
- B) $\frac{459\pi}{6}$
- C) $\frac{462\pi}{6}$
- D) $\frac{468\pi}{6}$

31. Orkhan Khalilzade

Two data Sets A and B have 56 values in total. The average of the two data sets is 193. Data set A contains 32 values with an average of 208. What is the average of data set B ?

32. Orkhan Khalilzade

Lines k and l are perpendicular. If the point $(4, m)$ and $(9, m + 3)$ lies on line k and lines k and l intersect at $(9, m + 3)$, which of the following can lie on line l ?

- A) $(m - 6, -4)$
- B) $(m + 9, 24)$
- C) $(21, m - 17)$
- D) $(24, m + 12)$

33. Orkhan Khalilzade

Data set A consists of 37 different values that have a minimum of 230, a maximum of 278, a mean of 244, and a standard deviation of 9. The values 230 and 278 are removed from the data set to create data set B . Which of the following statements is true?

- A) Both the mean and standard deviation of data set B are less than those in data set A .
- B) Both the mean and standard deviation of data set B are greater than those in data set A .
- C) The standard deviation of data set B is less than the standard deviation in data set A but the mean of both data sets is the same.
- D) The standard deviation of data set B is greater than the standard deviation in data set A but the mean of both data sets is the same.

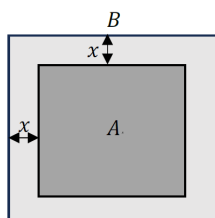
34. Orkhan Khalilzade

If $a > 0$, $x^2 + y^2 = a$ and $xy = a - 10$, what is $(x + y)^2$ in terms of a ?

- A) $a - 20$
- B) $2a - 10$
- C) $2a - 20$
- D) $3a - 20$

35. Orkhan Khalilzade

Rectangle A has a width of 50 feet and a length of 45 feet and is located inside rectangle B . A uniform distance x , in feet, separates the sides of rectangle A from rectangle B . The area of rectangle B is 2646 square feet, what is the value of x ?

**36.** Orkhan Khalilzade

$f(x) = 200(b)^x$. When x increases by 1, $f(x)$ increases by $c\%$. Which of the following expresses c in terms of b ?

- A) $c = \frac{200b}{100}$
- B) $c = \frac{2b}{100}$
- C) $c = (b + 1)100$
- D) $c = (b - 1)100$

37. Orkhan Khalilzade

A 34 pound dog eats two types of canned food: chicken and beef. The recommended amount of the chicken food is 1.25 cans per 16 pounds a dog weighs per day. The recommended amount of canned beef food is 0.85 cans per 23 pounds a dog weighs per day. If c is the number of cans of chicken food and b is the number of cans of beef food a 34 pound dog eats in a given day. Which equation describes all possible values of c and b ?

- A) $\frac{1.25}{16}c + \frac{0.85}{23}b = 34$
- B) $\frac{16}{1.25}c + \frac{23}{0.85}b = 34$
- C) $\frac{1.25}{16}b + \frac{0.85}{23}c = 34$
- D) $\frac{16}{1.25}b + \frac{23}{0.85}c = 34$

38. Orkhan Khalilzade

A quadratic function models the height, in feet, of an object above the ground in terms of time, in seconds, after the object is launched. The model indicates the object launched from the ground (0 feet) and reaches its maximum height of 576 feet above the ground 12 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 16 seconds after being launched?

- A) 384
- B) 432
- C) 480
- D) 512

39. Orkhan Khalilzade

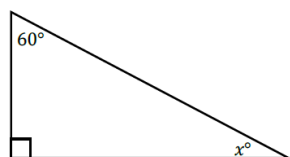
A small business owner budgets \$3,600 to purchase timber each year. The owner must purchase a minimum of 345 slats of timber each year to maintain the discounted pricing. If the owner pays \$7.25 to purchase each slat of pine pulpwood and \$24.75 to purchase each slat of pine sawtimber, what is the maximum number of slats of pine sawtimber the owner can purchase to stay within the budget and maintain the discounted pricing?

40. Orkhan Khalilzade

A 180 gram metal alloy is 50% aluminum. It contains a metal alloy that is 30% aluminum and a second metal alloy made up of 60% aluminum. What is the mass of the second metal alloy?

41. Orkhan Khalilzade

In the given figure, the hypotenuse is 64 what is the $\cos x$?



- A) $\frac{\sqrt{3}}{2}$
- B) $\frac{1}{2}$
- C) $\frac{2\sqrt{3}}{3}$
- D) 2

42. Orkhan Khalilzade

A quadratic function models the height, in feet, of an object above the ground in terms of time, in seconds, after the object is launched off an elevated surface. The model indicates that at a time of 9 seconds, the object is 294 feet above the ground. At a time of 13 seconds, the object is 310 feet above the ground. If the object was at a height of 24 feet when it was launched, what is the height, in feet, of the object above ground 18 seconds after being launched?

- A) 96
- B) 192
- C) 222
- D) 240

43. Orkhan Khalilzade

The table shows the distribution of 88 basketball players and their positions on a basketball club. Each player is categorized in one position.

Position	Frequency
Point guard	26
Shooting guard	16
Small forward	11
Power forward or center	35

If one of the 88 players is selected at random, the probability of selecting a player who is categorized as a power forward, given the player is not categorized as a shooting guard, is $\frac{3}{12}$. How many of these players are categorized as a center?

44. Orkhan Khalilzade

A rectangular pyramid has a square base with an area of 324 square meters. What is the surface area, in square meters, of one of the triangular faces if the rectangular pyramid has a volume of 4,320 cubic meters?

45. Orkhan Khalilzade

$n^{\frac{5}{3}} = k^{\frac{2}{5}}$ and $n^{4a+1} = k$, what is the value of a ?

46. Orkhan Khalilzade

$3x^2 + bx - 112 = (ax + m)(x - l)$. Given that a , m , and l are integers, which of the following must be true?

- A) m is a factor of b
- B) a is a factor of 112
- C) a is a factor of b
- D) l is a factor of 112

47. Orkhan Khalilzade

The perimeter of a right isosceles triangle is $76 + 76\sqrt{2}$. What is the length of the hypotenuse?

48. Orkhan Khalilzade

Sample	Percent in favor	Margin of error
<i>A</i>	67.83%	7.8%
<i>B</i>	72.94%	4.5%

The results of two random samples of votes for a proposition are given in the table. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample *A* is greater than the margin of error for sample *B*?

- A) Sample *A* had a larger sample size.
- B) Sample *A* had a smaller sample size.
- C) Sample *A* had a smaller number of votes that could not be recorded.
- D) Sample *A* had a lower percent of favorable responses.

49. Orkhan Khalilzade

360 is $p\%$ of 10. What is the value of p ?

50. Orkhan Khalilzade

The vertex of the function $f(x)$ is given as $(9, -13)$, and the point $(3, -4)$ lies on the parabola. If $g(x) = 4f(x - 2)$, what is the value of $g(0) - f(0)$?

51. Orkhan Khalilzade

$$-377x + k = 377x + k$$

If k is a constant, how many solutions does the given equation have?

- A) No solutions
- B) One solution
- C) Two solutions
- D) Infinite solutions

52. Orkhan Khalilzade

The density of a certain type of wood is 770 kilograms per cubic meter. A sample of this type of wood is in the shape of a cube, where each edge has a length of 0.7 meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- A) 264
- B) 377
- C) 1,571
- D) 2,245

53. Orkhan Khalilzade

The graph of the linear equation passes through the points $(-21, -\frac{89}{2})$ and $(12, \frac{65}{2})$. The graph also passes through the point $(\frac{27}{4}, a)$. What is the value of a ?

54. Orkhan Khalilzade

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{AB}$ is 456 millimeters. If the $\cos C = \frac{7}{25}$, what is the length, in millimeters, of $\overline{BC}$?

55. Orkhan Khalilzade

In triangle ABC and triangle DEF , sides BC and EF each have a side length of 37 inches, and angles B and E each have an angle measure of 63° . Which of the following additional pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?

- I. The measure of angles A and C are equal.
- II. The length of sides AC and DF are equal.
- III. The measures of angles A and D are equal.

- A) I is sufficient, but II and III are not.
- B) II is sufficient, but I and III are not.
- C) III is sufficient, but I and II are not.
- D) II and III are sufficient, but I is not.

56. Orkhan Khalilzade

$$77x^2 + (7j + 11k)x + jk = 0$$

In the given equation, j and k are positive constants. The product of the solutions to the given equation is ajk , where a is a constant. What is the value of a ?

57. Orkhan Khalilzade

The number j is 2,400% greater than the number k . The number k is 350% greater than 8. What is the value of j ?

58. Orkhan Khalilzade

A square is inscribed in a circle with the equation $x^2 + y^2 - 8x + 10y = 27$. What is the area of the square?

59. Orkhan Khalilzade

The equation below relates distinct positive real numbers a, b and c . Which equation correctly expresses c in terms of a and b ?

$$a = 19b \cdot \sqrt[5]{\left(\frac{c}{20}\right)^4}$$

- A) $\frac{20}{19b}(a)^{\frac{5}{4}}$
- B) $\frac{20}{19b}(a)^{\frac{4}{5}}$
- C) $20\left(\frac{a}{19b}\right)^{\frac{5}{4}}$
- D) $20\left(\frac{a}{19b}\right)^{\frac{4}{5}}$

60. Orkhan Khalilzade

Triangle ABC is inscribed in a circle with a radius of 85. The length of AC is 170 and the length of BC is 168. What is the value of $\frac{AB}{BC}$?

61. Orkhan Khalilzade

For the exponential function f , the value of $f(2)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

- A) $f(x) = 729(3)^{x+2}$
- B) $f(x) = 6,561(3)^x$
- C) $f(x) = 177,147(3)^{x-3}$
- D) $f(x) = 59,049(3)^{x-2}$

62. Orkhan Khalilzade

Two numbers r and s are each greater than zero, and the fourth root of r is equal to the seventh root of s . For what value of x is r^{4x-5} equal to s ?

63. Orkhan Khalilzade

The given equation $ax^2 + 98x + c$ has at least 1 real root and a factor of $kx + j$. What is the greatest possible value of ac ?

64. Orkhan Khalilzade

$$7kx + 13my = 20.4$$

$$6kx + 5my = -48$$

In the given system of equations, k and m are constants. The system has a solution of $(\frac{66}{43}, y)$. What is the value of k ?

65. Orkhan Khalilzade

x	$g(x)$
4	9
2	0

The table gives two values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)+5}{x+8}$ and f is a linear function. What is the y -coordinate of the y -intercept of the graph $y = f(x)$ in the xy -plane?

66. Orkhan Khalilzade

What is the value of $\sin(x^\circ)(\cos 90 - x^\circ) + \sin(90 - x^\circ)(\cos x^\circ)$?

- A) 0.5
- B) 1
- C) 2
- D) 90

67. Orkhan Khalilzade

There are red, blue, green, and black marbles in a bag. There are 26 black marbles, 8 green marbles, and 26 blue and red marbles. Given that a randomly selected marble is not black, the probability of selecting a red marble is $\frac{5}{17}$. What is the number of blue marbles in the bag?

68. Orkhan Khalilzade

A gardener is purchasing topsoil for a new garden. The garden is in the shape of a rectangular prism. The dimensions of the garden are 54 feet by 192 feet by 2 feet. If it costs \$96 per cubic yard of topsoil, how much does the gardener need to spend in dollars? (1 yard = 3 feet)

69. Orkhan Khalilzade

In the xy -plane, which of the following points lies on a circle with the equation $(x + k)^2 + (y - j)^2 = 841$?

- A) $(k + 20, j + 21)$
- B) $(k - 20, j - 21)$
- C) $(-k + 21, j + 20)$
- D) $(-k - 21, -j - 20)$

70. Orkhan Khalilzade

$$4 - 4 \left| \frac{7}{8} + \frac{4}{5}x \right| = \frac{57}{12} + k$$

Which of the following inequalities gives all possible values of k where the equation below has no solution?

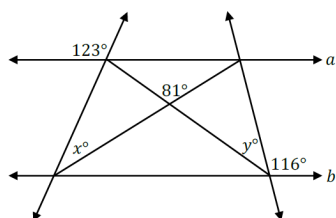
- A) $k < -\frac{3}{4}$
- B) $k > -\frac{3}{4}$
- C) $k \leq \frac{3}{4}$
- D) $k \geq \frac{3}{4}$

71. Orkhan Khalilzade

A table that weighs 348 pounds puts equal amounts of pressure on each of its 4 identical legs. If the pressure that each leg exerts on the floor is 12 pounds per square inch, what is the area, in square inches, of the bottom of each leg that is touching the floor?

72. Orkhan Khalilzade

In the figure below, lines a and b are parallel. What is the value of $x + y$?

**73.** Orkhan Khalilzade

The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

- I. $f(x) = 232(0.4)^{x+2}$
 II. $g(x) = 232(0.4)(0.4)(0.4)^{x-2}$

- A) I only
 B) II only
 C) I and II
 D) Neither I nor II

74. Orkhan Khalilzade

An equilateral triangle with side lengths of k is inscribed in a circle. Which of the following defines the radius, r , in terms of k ?

- A) $\frac{k}{2}$
 B) $\frac{k\sqrt{3}}{6}$
 C) $\frac{k\sqrt{3}}{3}$
 D) $\frac{k\sqrt{3}}{2}$

75. Orkhan Khalilzade

The function f is defined by $f(x) = 3^x$. The function g is an increasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points (a, j) and (b, k) , where $j < k$. When $g(x) > f(x)$, which of the following must be true?

- A) $x > k$
 B) $x < j$
 C) $x < j$ or $x > b$
 D) $a < x < b$

76. Orkhan Khalilzade

If $\frac{2a}{b} = 4.25$ and $\frac{a}{bm} = 8.5$, what is the value of n ?

77. Orkhan Khalilzade

An object's speed is increasing at a rate of 4.80 meters per second squared. What is the rate in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1609 meters.)

78. Orkhan Khalilzade

An acceptable noise criterion rating for the background noise in a school classroom is 30. For a noise criterion rating of 30, the equation $y = 19(0.978)^{x-50} + 37$ gives the estimated sound pressure level, y in decibels, as a function of the octave band center frequency, x , in hertz, where $x \geq 50$. Which of the following is the best interpretation of 37 in this context?

- A) 37 is 19 less than the estimated sound pressure level, in decibels, at an octave band frequency of 50.
- B) 37 is 19 less than the estimated sound pressure level, in decibels, at an octave band frequency of 0.
- C) 37 is the estimated sound pressure level, in decibels, at an octave band frequency of 50.
- D) 37 is the estimated sound pressure level, in decibels, at an octave band frequency of 0.

79. Orkhan Khalilzade

The graph of the quadratic function $y = f(x)$ in the xy -plane intersects the x -axis when $x = 74$ and $x = k$, where k is a constant. The maximum value of $y = f(x)$ occurs at the point $(17, m)$, where m is a constant. What is the value of k ?

80. Orkhan Khalilzade

x	$f(x)$
-9	155
-3	227
3	155

Three points on the quadratic function $f(x)$ are given in the table. If $g(x) = f(x+4)$, what is the y -intercept of the function $g(x)$?

81. Orkhan Khalilzade

Triangle ABC is an equilateral triangle with a height of 30 meters. What is the perimeter of triangle ABC in meters?

- A) $60\sqrt{3}$
- B) 90
- C) $90\sqrt{3}$
- D) $120\sqrt{3}$

82. Orkhan Khalilzade

$$-19(6x-4)^2 + 6(6x-3)^2$$

The given expression can be rewritten as

$\frac{a}{9}x^2 + \frac{b}{9}x + \frac{c}{9}$, where a , b , and c are constants. What is the value of $a + b + c$?

83. Orkhan Khalilzade

A rectangular area consists of 3,179 equal squares where each square has an area of k . If the length of the rectangular area is 2.75 times the width and the width is equal to $x\sqrt{k}$, what is the value of x ?

84. Orkhan Khalilzade

$$7x - 9y = 39$$

$$35x - 45y = 195$$

For any real number r , which of the following points lies on the graph of both equations?

- A) $(3r, \frac{7r-13}{3})$
- B) $(\frac{7r+13}{3}, 3r)$
- C) $(-\frac{7r}{9} + 39, \frac{7r}{9} + 195)$
- D) $(\frac{r}{5} + 39, -\frac{r}{5} + 195)$

85. Orkhan Khalilzade

$$f(x) = x^2 + 5x - 6$$

$$g(x) = x^2 + 2x - 35$$

The quadratic function $f(x)$ has 2 solutions j and k , where $j < k$. The quadratic function $g(x)$ has 2 solutions l and m , where $l < m$. The quadratic function $h(x) = x^2 + 7x + c$ has solutions of $k + l$ and $m + j$, and can be rewritten as $(x + a)(x + b)$, what is the value of c ?

86. Orkhan Khalilzade

For the exponential function $f(x) = 16^{x+4}$, which of the following answer choices expresses the minimum value as either a constant or a coefficient when $x \geq 0$?

A) $f(x) = 64(16)^x$

B) $f(x) = 65,536(16)^x$

C) $f(x) = 1,048,576(16)^x$

D) $f(x) = \frac{1}{65,356}(16)^x$

87. Orkhan Khalilzade

For the given parabola $ax^2 + bx + c$, a , b , and c are constants. The x -intercepts of the parabola are at $(14, 0)$ and $(k, 0)$. If $f(27) = f(11)$, what is the value of k ?

88. Orkhan Khalilzade

A quadratic function models a projectile's height, in meters, above the ground in terms of in terms of time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of a meters and reached a maximum height of 134 meters 6 seconds after it was launched. How many seconds after launch does the model estimate it will return to a height of a meters?

A) 6

B) 8

C) 9

D) 12

89. Orkhan Khalilzade

A 100 gram metal alloy is 40% silver. It contains a metal alloy that is 65% silver and a second metal alloy made up of 25% silver. What is the mass of the first metal alloy?

90. Orkhan Khalilzade

Joshua Tree National Park has an area of 1,242 square miles. What is the area, in square yards, of Joshua Tree National Park? (1 mile = 1,760 yards).

A) $\frac{1,242}{1,760}$

B) $\frac{1,242}{(1,760)^2}$

C) $1,242(1,760)$

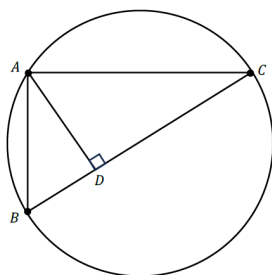
D) $1,242(1,760)^2$

91. Orkhan Khalilzade

There are two ranches. Ranch A and ranch B . The ratio of male horses to female horses is 1: 15 for the total number of horses on ranch A and B . The ratio of female horses on ranch A to ranch B is 7 : 17. There are 168 horses on ranch A and 344 horses on Ranch B . How many more male horses are there on ranch A than on ranch B ?

92. Orkhan Khalilzade

In the given figure, BC is the diameter of the circle. If the length of BC is equal to 132 and the length of AB is equal to $\sqrt{363}$, what is the value of $\frac{BC}{BD}$?



Note: Figure not drawn to scale.

93. Orkhan Khalilzade

A certain bacterial colony currently has 5,000 spores. If the number of spores in this colony quadruples every 33 hours, which of the following functions N gives the number of spores in the colony d days from now?

- A) $N(d) = 5,000(4)^{\frac{11}{8}d}$
- B) $N(d) = 5,000(4)^{\frac{8}{11}d}$
- C) $N(d) = 5,000(4)^{\frac{5}{4}d}$
- D) $N(d) = 5,000(4)^{\frac{4}{5}d}$

94. Orkhan Khalilzade

$$\sqrt[7]{a^5} = \sqrt{b^5}$$

$$a^{3x-3} = b^3$$

In the given equations, a and b are constants, $a > 1$ and $b > 1$. What is the value of x ?

95. Orkhan Khalilzade

a is 3700% greater than b . a is also 5% less than c . What percent greater than b is c ?

96. Orkhan Khalilzade

The proposal for a development project was included on a city's election ballots. A local TV station reported that 6 times as many people voted in favor of the proposal as people who voted against it. If 1,770 more people voted in favor of the proposal than those who voted against the proposal. How many people voted against the proposal?